Supplementary Materials for "Membership Inference Attacks Beyond Overfitting"

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No Institute Given

The content herein provides detailed experimental setup.

1 Experimental setup

The experiments were conducted on a system running Windows 11 Home OS, equipped with a 12$^{\text{th}}$ Gen Intel®Core™i7-12700 12-core CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4080 with 16 GB GPU. Our implementation is based on the code provided by [4]. Our code is available at `https://anonymous.4open.science/r/mia_analysis-31C6`.

1.1 Datasets

We used two benchmark datasets commonly employed in the literature on MIAs, described in what follows:

- *Purchase100* [12] is a dataset comprising 197,324 shopping records, each represented by 600 binary features indicating whether a specific item was purchased. The goal is to classify customers into one of 100 distinct classes based on their purchasing patterns. This helps understand consumer behavior for targeted marketing and personalized recommendations. We split the data into 80% for training and 20% for testing.
- *CIFAR-10* [8] is an image classification dataset with 60,000 color images in 10 classes, such as airplanes, automobiles, birds, and cats. Each image is 32x32 pixels with three color channels (RGB). This dataset is widely used for training and evaluating image recognition models. We used 50,000 samples for training and 10,000 samples for testing.

1.2 Models

We employed different models tailored to each dataset.

For Purchase100, we used a fully-connected network (FCN) as in [11]. This model classifies purchasing patterns into 100 classes with a feature extractor comprising an input layer with 600 nodes, four fully connected hidden layers (resp. with 1024, 512, 256, and 128 nodes) with tanh activation functions, followed by a classification layer. The architecture has approximately 1,320,000 parameters.

For CIFAR-10, we employed two convolutional neural network (CNN) architectures: DenseNet-12 [6] and ResNet-18 [5]. DenseNet-12, known for its densely connected layers, promotes feature reuse and improves gradient flow. ResNet-18, a residual network, aids in training deeper networks by introducing shortcut connections to mitigate the vanishing gradient problem. DenseNet-12 has about 770,000 parameters, whereas ResNet-18 has about 11,170,000 parameters.

1.3 Membership inference attacks

We used two black-box membership inference attacks implemented in TensorFlow Privacy [1]: the loss-based attack by [15] and the entropy-based attack by [10].

1.4 Evaluation metrics

We evaluated the utility of the ML models using their accuracy (Acc). For MIAs, we used AUC to quantify the trade-off between true positives and false positives for member and non-member data points. We also used the attacker's advantage [15], defined as the maximum absolute difference between the attacker's true positive rate (TPR) and false positive rate (FPR) across all thresholds. Additionally, we identified the most vulnerable samples using true positives (TP) at a low false positive rate (FPR), as suggested in [3]; these are the samples that are most neatly detectable by the attacker.

1.5 Training settings

We trained the FCN model on Purchase100 for 200 epochs using a learning rate 5e-4 and a batch size 32, with the cross-entropy loss function and the Adam optimizer. For CIFAR-10, DenseNet-12 and ResNet-18 were trained for 100 epochs with an initial learning rate 0.1 and a batch size 64, using the cross-entropy loss and SGD optimizer. The learning rate was reduced by a factor of 10 at the 30th, 60th, and 90th epochs.

1.6 Defense settings

We considered the following defenses: early stopping [13], L2-regularization [12], regularization and dropout (RegDrop) [2], label smoothing (LS) [14], and DP-SGD [1].

For early stopping, we halted training if no improvement was observed on the validation set for 10 consecutive epochs. For L2-regularization, we tested different λ values in $\{5e-4, 1e-3, 5e-3\}$. For RegDrop, we fixed $\lambda = 5e-4$ and used dropout ratios in $\{0.25, 0.5\}$. For label smoothing, we followed [7], training Purchase100-FCN with a smoothing intensity 0.03 and CIFAR-10 DenseNet-12 with a smooting intensity 0.01.

For DP-SGD, we used PyTorch Opacus [9], setting the microbatch size to 1. Purchase100-FCN was trained with a noise multiplier 1.0, a norm clipping bound 1.0, and for 200 epochs. CIFAR-10 DenseNet-12 was trained with similar settings, except for the noise multiplier, which was set to 0.5.

[1] `https://www.tensorflow.org/responsible_ai/privacy/guide`

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